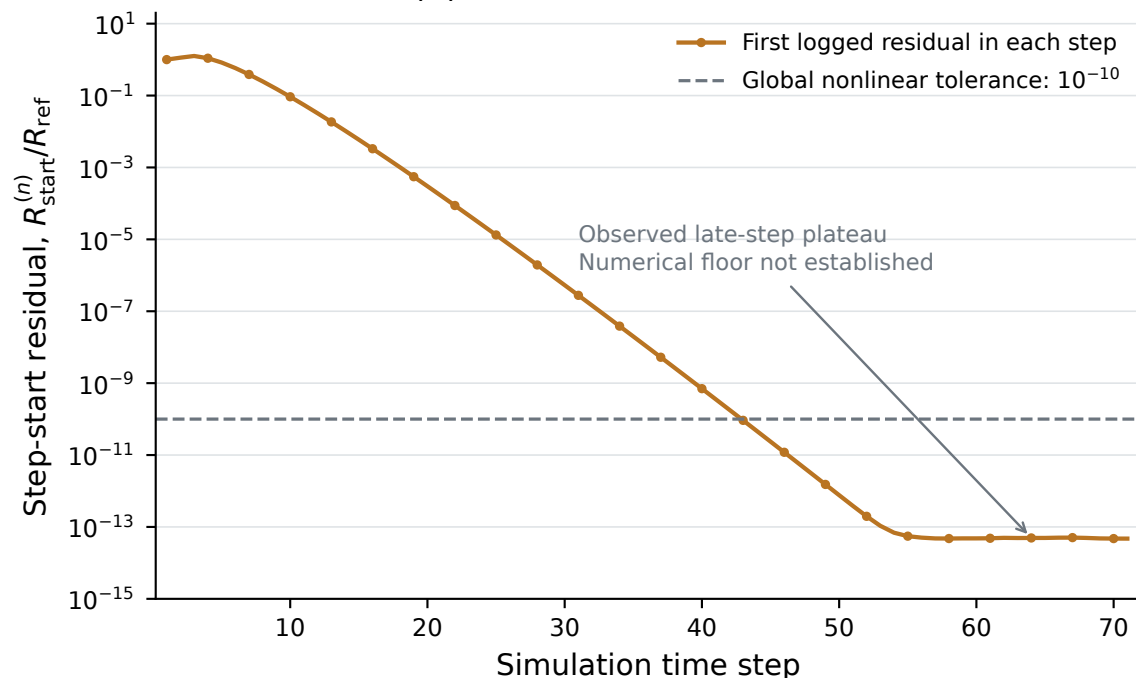


Nonlinear Residual Evolution and Within-Step Reduction

$R_{\text{ref}} = R^{(1,1)}$ is fixed for the entire run; $R_{\text{norm}}^{(n,k)} = R^{(n,k)} / R_{\text{ref}}$ (k is 1-based).

Start / last = first / last logged records in each step. R is the nonlinear solver's scaled residual measure.

(A) Global transient evolution



Stopping rule: at least 2 iterations, then global OR step-relative normalized residual $\leq 10^{-10}$.

Steps 4 and 23 use one fewer iteration than the preceding step; tolerance is met, with no preconditioner rebuild.

A lower nonlinear residual indicates better satisfaction of the discretized nonlinear equations within that time step.

Residual magnitude is not a CFD solution-error estimate. The late plateau is observed, not a verified floating-point floor.

(B) Nonlinear reduction within each step

